

7-4 Test Results and Analysis

Analysis is based on the automated test inventory in the repository, PHPUnit execution, and acceptance scenarios on SPU demo data.

1) Automated Inventory Summary

Metric	Value
Total test cases (list-tests)	141
Unit	50
Feature / System	91
Test files	20 files (Unit + Feature)

2) Coverage by Main Test Files

File	Type	Verification Focus
ProjectProposalControllerTest	Feature	Submit / approve / delete proposals and tenant isolation
TrackControllerTest / TrackServiceTest	Feature + Unit	Tracks, progress, and prerequisites
XmlImport* / XmlRegistrationTest	Feature + Unit	Import and registration matching
CommitteeControllerTest	Feature	Committee management and conflicts
GeneticScheduler* / Fitness* / Evolutionary*	Unit	Scheduling engine and constraints
ScheduleControllerTest / ScheduleApproval*	Feature + Unit	Generation and approval
AIIdeation* / AITask*	Unit + Feature	AI services

3) Results Analysis

- **Passed cases:** cover critical paths (XML, proposals, tracks, committees, scheduling).
- **Failures / errors:** fixed immediately then re-validated with Regression; logged when they appear in full runs.
- **Remaining defects:** no critical security or tenant-isolation defects are accepted before delivery.
- **Execution time:** depends on machine and test DB; recorded from PHPUnit output for formal runs.

Official documentation command: `cd "backend last" && php vendor/bin/phpunit --testdox`